

座號：

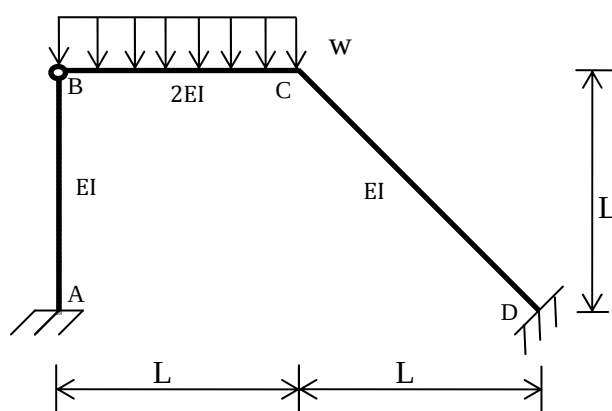
(二)不必抄題，作答時請將試題題號及答案依照順序寫在試卷上，於本試題上作答者，不予計分。

The diagram shows a truss structure with 12 joints labeled a through l. The structure is supported by a pin support at joint a and a roller support at joint i. The dimensions are as follows: vertical height segments are 10 m (from a to b, b to c, and j to i), and horizontal segments are 5 m (from a to b, b to d, d to e, e to f, and f to g). The total horizontal span is 30 m. The loads are: 5 kN downward at joint c, 10 kN downward at joint d, 10 kN downward at joint e, 10 kN downward at joint f, and 5 kN downward at joint g. The truss members connect the following joints: (a,b), (b,c), (c,d), (d,e), (e,f), (f,g), (g,h), (h,i), (i,j), (j,k), (k,l), (l,a), (b,k), (d,k), (e,j), (f,j), and (g,i).

(請接背面)

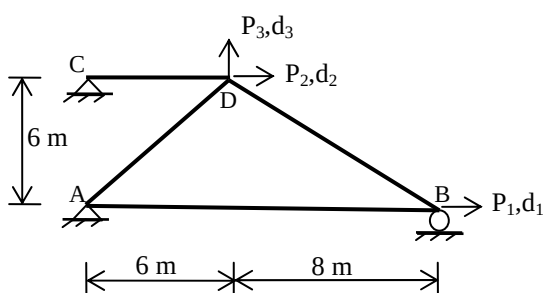
等 別：高等考試
類 科：結構工程技師
科 目：結構學

三、所示剛架結構 A、D 為固接支承，B 點為鉸接。桿件 BC 承受垂直均佈載重 w ，如圖三所示。各桿件之 E 均相同。桿件 AB、CD 面積慣性矩為 I ，桿件 BC 則為 $2I$ 。請以傾角變位法分析各桿件端點彎矩，並繪桿件 BC 剪力圖與彎矩圖。使用其他方法不予給分。(25 分)



圖三

四、如圖四所示桁架結構，請應用直接勁度法組成對應位移(d_1 、 d_2 、 d_3)之勁度矩陣。若所得位移(d_1 、 d_2 、 d_3)為(0.1277, 0.1743, 0.7055)mm，求所對應施加外力(P_1 、 P_2 、 P_3)值，並求桿件 AD、CD 之內力。所有桿件 $E=200$ GPa， $A=10 \times 10^3$ mm²。使用其他方法不予給分。(25 分)



圖四